

UYAR, Russian Federation

WMO: 295760

Lat: 55.8192N

Lon: 94.3143E

Elev: 376

StdP: 96.89

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -38.1	(c) -34.1	(d) -40.4	(e) 0.1	(f) -37.8	(g) -36.8	(h) 0.1	(i) -33.8	(j) 11.5	(k) -6.8	(l) 10.3	(m) -7.8	(n) 1.7	(o) 40	(p) 0.606

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 9.4	(c) 28.4	(d) 18.3	(e) 26.7	(f) 17.5	(g) 25.0	(h) 16.9	(i) 19.6	(j) 25.7	(k) 18.7	(l) 24.4	(m) 17.9	(n) 23.1	(o) 2.3	(p) 60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.6	13.2	21.8	16.7	12.4	20.8	15.7	11.7	20.4	57.8	25.8	54.8	24.5	52.1	23.1	21.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 9.5	(b) 6.4	(c) 5.7	(e) -38.6	(f) 31.8	(g) 4.2	(h) 1.0	(i) -41.6	(j) 32.6	(k) -44.1	(l) 33.2	(m) -46.5	(n) 33.8	(o) -49.5	(p) 34.6		
(a) 9.5	(b) 6.4	(c) 5.7	(e) -38.1	(f) 20.9	(g) 4.0	(h) 0.8	(i) -41.0	(j) 21.5	(k) -43.3	(l) 21.9	(m) -45.6	(n) 22.4	(o) -48.5	(p) 23.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	0.9	-17.9	-13.2	-5.9	2.8	9.2	16.1	18.0	15.8	7.9	0.3	-10.3	-13.6		
	DBStd	13.82	9.72	7.67	6.83	4.73	5.39	4.25	2.78	3.64	4.43	5.33	8.64	8.56		
	HDD10.0	4052	866	649	492	220	83	5	0	2	92	301	610	732		
	HDD18.3	6454	1124	882	750	466	287	89	40	95	312	559	860	991		
	CDD10.0	713	0	0	0	4	59	189	247	183	29	1	0	0		
	CDD18.3	73	0	0	0	0	3	23	29	17	0	0	0	0		
	CDH23.3	736	0	0	0	2	51	279	238	161	6	0	0	0		
	CDH26.7	138	0	0	0	0	8	57	40	33	0	0	0	0		
(14) Wind	WSAvg	2.4	2.3	2.3	2.6	2.9	2.8	2.1	1.9	1.8	2.0	2.4	2.6	2.6		
(15) Precipitation	PrecAvg	481	17	16	17	31	42	51	64	89	48	38	35	30		
	PrecMax	580	31	26	56	64	89	81	161	193	72	59	57	59		
	PrecMin	344	5	8	4	7	18	13	8	45	7	8	18	9		
	PrecStd	68	7	6	13	15	21	16	39	38	20	15	10	13		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.7	3.0	9.8	19.9	27.7	30.6	30.5	29.6	23.8	17.0	8.5	2.6		
		MCWB	-1.2	-0.6	3.8	9.1	15.4	18.3	19.7	19.6	14.8	9.2	4.6	-0.2		
	2%	DB	-0.4	0.7	7.1	16.3	24.4	28.3	27.8	27.1	20.0	12.6	5.2	0.1		
		MCWB	-2.8	-2.3	2.6	7.9	13.7	17.8	18.8	18.5	13.6	7.4	2.0	-2.1		
	5%	DB	-2.6	-0.9	5.0	13.3	21.3	26.5	26.2	24.8	17.3	10.0	3.1	-1.5		
		MCWB	-4.5	-3.2	1.1	6.6	12.1	17.1	18.0	17.3	12.3	6.1	0.4	-3.0		
	10%	DB	-4.9	-2.5	3.0	11.0	19.0	24.7	24.3	22.7	14.7	7.8	0.8	-3.1		
		MCWB	-6.4	-4.3	-0.2	5.4	11.3	16.3	17.4	16.6	10.8	4.8	-0.9	-4.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-0.7	-0.4	4.6	10.3	16.2	20.0	20.7	20.7	15.1	9.7	5.0	0.3		
		MCDB	1.2	2.3	8.5	17.6	25.7	27.3	27.2	27.8	21.9	14.5	8.0	2.4		
	2%	WB	-2.9	-1.8	2.8	8.7	14.5	18.9	19.7	19.2	14.0	8.0	2.3	-1.8		
		MCDB	-0.8	0.2	6.7	14.9	22.7	26.2	25.5	25.1	19.4	11.7	5.0	-0.2		
	5%	WB	-4.4	-3.0	1.4	7.1	13.2	18.0	18.9	18.2	12.7	6.6	0.7	-3.0		
		MCDB	-2.4	-1.0	4.7	12.6	19.8	24.6	24.2	23.2	16.5	9.5	2.9	-1.4		
	10%	WB	-6.4	-4.4	0.0	5.6	11.7	17.0	18.1	17.1	11.5	4.9	-0.9	-4.5		
		MCDB	-4.8	-2.4	2.6	10.6	17.8	23.1	22.9	21.8	14.3	7.6	0.7	-3.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	7.3	8.4	9.0	9.2	10.5	10.9	9.4	9.7	8.6	7.2	7.0	6.8		
		MCDBR	8.8	8.5	10.1	13.2	14.7	14.7	12.7	13.2	13.6	12.0	7.7	7.8		
		MCWBR	7.1	6.6	6.7	7.5	7.2	6.0	5.0	5.6	7.1	7.4	5.5	6.6		
		MCDBR	8.2	8.0	9.3	12.2	12.6	13.0	10.4	11.7	12.4	10.4	7.6	7.5		
	5% WB	MCWBR	6.6	6.4	6.4	7.5	6.5	5.5	4.5	5.3	6.7	6.8	5.8	6.6		
(40) Clear-Sky Solar Irradiance	taub		0.275	0.308	0.351	0.387	0.406	0.401	0.420	0.402	0.360	0.340	0.308	0.276		
	taud		2.034	2.016	1.985	2.144	2.201	2.307	2.271	2.288	2.360	2.262	2.082	1.977		
	Ebn at Noon		601	728	788	826	834	843	816	805	792	698	547	480		
	Edh at Noon		65	99	132	131	132	120	123	113	91	78	63	54		
(44) All-Sky Solar Radiation	RadAvg		0.81	1.82	3.24	4.34	4.97	5.93	5.35	4.33	2.79	1.53	0.77	0.50		
	RadStd		0.08	0.11	0.22	0.25	0.49	0.50	0.39	0.37	0.33	0.12	0.09	0.05		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB		1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (14 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page